

Assignment 2 Report: Sanskrit-to-English Neural Machine Translation

Course: Natural Language Understanding

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GitHub: <https://github.com/rishikeshwagh/NLUAssignment2.git>

Introduction

Neural Machine Translation (NMT) has significantly improved the quality of automatic language translation by learning contextual representations directly from bilingual corpora. Sanskrit, being a morphologically rich and low-resource language, presents several challenges including complex grammatical structures, rich inflections, and flexible word order. The objective of this project is to develop a custom Sequence-to-Sequence (Seq2Seq) Neural Machine Translation model capable of translating Sanskrit sentences into English. The implemented model employs an Encoder-Decoder architecture with an Attention mechanism, allowing the decoder to focus on relevant source words while generating each target token. The complete implementation was developed using PyTorch and evaluated using BLEU and BERT Score metrics.

Dataset

The provided training, development and test datasets consist of aligned Sanskrit-English sentence pairs. Source IDs are preserved for prediction generation.

The dataset comprises **10,000 training samples**, **1,000 development samples**, and **1,000 test samples**. Each Sanskrit sentence is paired with its corresponding English translation using a common **Source ID**. The training set was used for model learning, the development set for evaluation, and the test set for generating the final translation predictions.

Before training the model, the dataset underwent several preprocessing steps to ensure consistency and efficient learning. The Sanskrit and English sentences were first **tokenized** into individual words or tokens. Separate **vocabularies** were then constructed for both source and target languages.

To facilitate sequence modeling, special tokens such as **Start of Sequence (SOS)**, **End of Sequence (EOS)**, **Padding (PAD)**, and **Unknown (UNK)** were introduced. Since sentence lengths vary across the dataset, **sequence padding** was applied to create fixed-length inputs within each batch. Finally, the processed data were organized into **mini-batches**, enabling efficient training and better utilization of computational resources during model optimization.

Architecture

A custom Seq2Seq Neural Machine Translation model was implemented. The architecture consists of: Encoder Network Decoder Network Attention Mechanism Greedy Decoder The encoder converts Sanskrit tokens into contextual hidden representations. During decoding, the attention mechanism dynamically identifies the most relevant encoder outputs for generating each English word. Greedy decoding is used during inference. Beam Search was not implemented because the objective of this assignment was to develop a custom baseline Seq2Seq model.

Training Procedure

The following steps were performed during training:

Dataset preprocessing, Vocabulary construction, Tokenization, Padding, Mini-batch creation, Teacher-forced sequence prediction, Cross Entropy Loss optimization, Adam optimizer, Model checkpointing

Hyperparameters and Experimental Setup

The proposed Seq2Seq Neural Machine Translation model was trained for **10 epochs** using the **Adam optimizer** with the **Cross Entropy Loss** function. During inference, the model employed **Greedy Decoding**, while **Beam Search was not implemented** in the current work. The final model consists of **17,270,240 trainable parameters**, providing sufficient capacity to learn contextual representations for Sanskrit-to-English translation. This configuration served as the baseline implementation for evaluating translation performance on the provided dataset

Results

The developed Seq2Seq Neural Machine Translation model achieved a **BLEU score of 0.0077** and a **Rescaled BERTScore (F1) of 0.0679** on the development dataset. The model consists of **17,270,240 trainable parameters** and completed inference on the private test dataset in approximately **4.00 seconds**. These results indicate that while the model successfully learned the training data and generated translations efficiently, there is significant scope for improving translation quality through techniques such as Beam Search, Transformer-based architectures, and larger parallel corpora.

Translation Examples

Source ID	Sanskrit Sentence	Reference Translation	Predicted Translation	Error Analysis	Outcome
1	एक्लिप्स् इति प्रोग्रामर् कृते दोषान्वेषणे अपि साहाय्यं करोति।	Eclipse also helps the programmer to find out errors.	In this tutorial, we will to	Prediction differs significantly from the reference translation.	Incorrect
2	विश्वासकारणादेव समभाषि मया वचः। इति यथा शास्त्रे लिखितं तथैवास्माभिरपि विश्वासजनकम् आत्मानं प्राप्य विश्वासः क्रियते तस्माच्च वचांसि भाष्यन्ते।	"We having the same spirit of faith, according as it is written, I believed, and therefore have I spoken; we also believe, and therefore speak;"	"And I think that they had to a new in the name is an error	Meaning partially preserved but wording differs.	Partial
3	तदा, तत्स्वयं ड्रैवर निमित्तम् अन्वेष्ट्यति। अहं 'Cancel' इत्यस्योपरि नुदामि।	Then it will automatically begin searching for drivers. I will click on Cancel.	I click on this tutorial	Meaning partially preserved but wording differs.	Partial
4	सर्वेभ्यः इटरेशन्-अर्थम्, iterator इतीदं प्रत्येकस्मै इण्डेक्स्-वेल्फू-इत्यस्मै सेट् क्रियते। 1,1 पश्चात् 1,2 एवम्...	The iterator will be set to each of the indices values for each iteration- say, 1 comma 1, then 1 comma 2 and so on.	The video available at the Spoken Tutorial project, watch to be a new	Meaning partially preserved but wording differs.	Partial
5	अपरं द्वितीयमुद्रायां तेन मोचितायां द्वितीयस्य प्राणिन	"And when he had opened the second seal, I heard the	"And they cried I had unto them that he was a ship,	Meaning partially preserved but wording differs.	Partial

	आगत्य पश्येति वाक् मया श्रुता।	second beast say, Come and see."			
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Training Analysis

The attention mechanism improves alignment between source and target tokens; however, limited training data and greedy decoding reduce translation quality on complex sentences. Future work includes beam search and Transformer architectures.

Appendix: Final Evaluation Metrics

Metric	Value
BLEU Score	0.0077
Rescaled BERTScore (F1)	0.0679
Inference Time (Private Test)	4.00 seconds
Pretrained Models	None for translation model; RoBERTa-Large used only by the bert-score library for evaluation.
Beam Search	Not implemented. Greedy decoding used during inference.

Experimental Observations

Training loss decreased steadily while validation loss stabilized over epochs, indicating that the model learned meaningful source-target alignments. The low BLEU (0.0077) and BERTScore (0.0679) indicate that the custom Seq2Seq model struggled to generalize on the development set due to the limited dataset size, Sanskrit's rich morphology, and greedy decoding. Despite the modest translation quality, the model successfully generated predictions for the complete test set in approximately 4 seconds.

Training and Validation Loss Curve



Conclusion

A complete **Sanskrit-to-English Neural Machine Translation (NMT)** system was successfully developed using a custom **Sequence-to-Sequence (Seq2Seq)** architecture with an **Attention mechanism**. The final model consists of **17,270,240 trainable parameters** and successfully generated translations for the complete test dataset in approximately **4.00 seconds**. During evaluation on the development dataset, the model achieved a **BLEU Score of 0.0077** and a **Rescaled BERTScore (F1) of 0.0679**. Although the obtained translation quality is modest, primarily due to the challenges of low-resource Sanskrit translation, rich morphological structures, and the use of greedy decoding, the implemented system establishes a **fully reproducible baseline NMT model**. Future work will focus on improving translation quality through **Beam Search decoding**, **Transformer-based architectures**, **larger parallel Sanskrit-English corpora**, and **advanced tokenization techniques** such as **Byte Pair Encoding (BPE)**.